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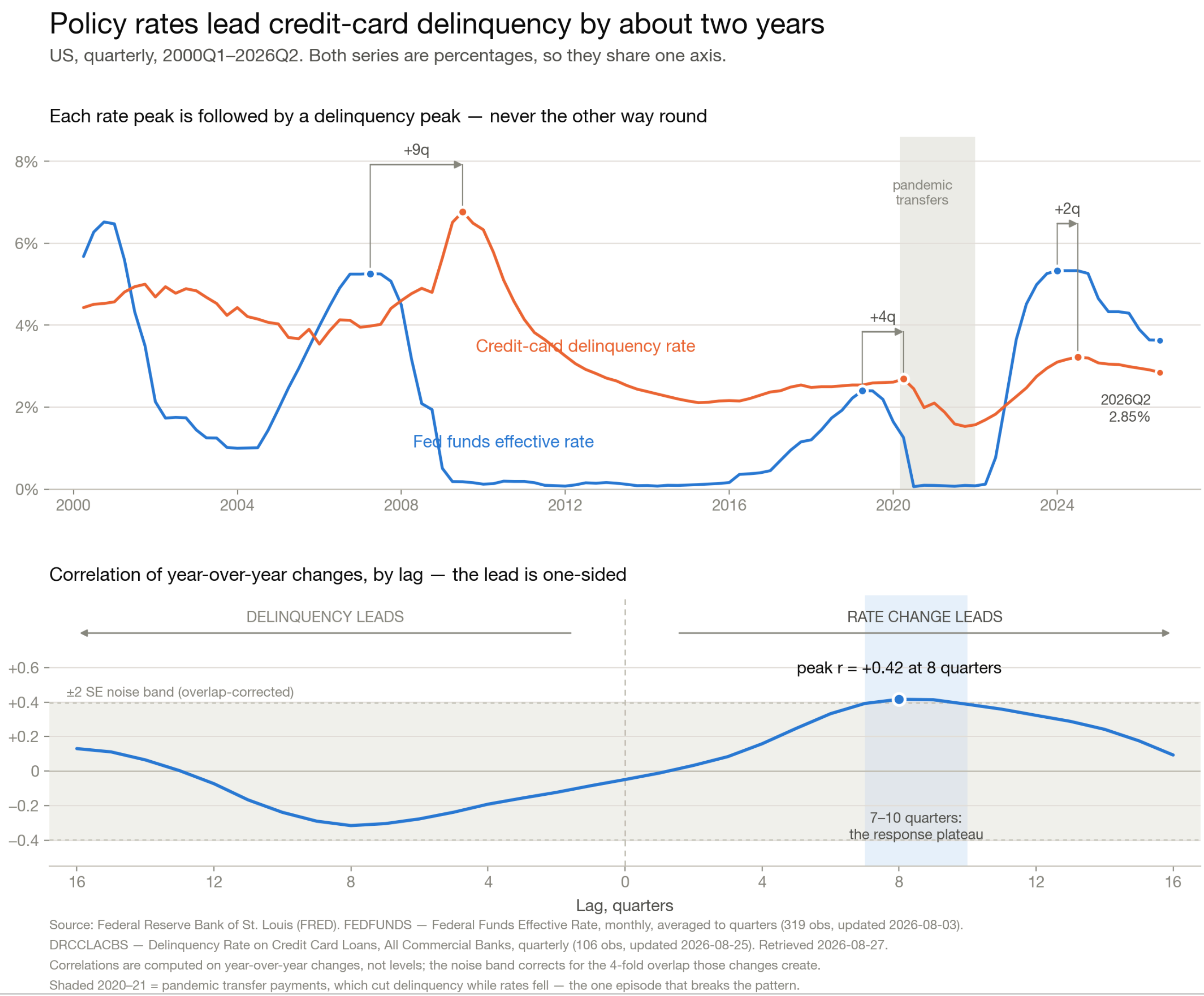
Rates lead card delinquency by about two years



Fed funds effective rate and credit-card delinquency rate, 2000Q1–2026Q2, with the cross-correlation of their year-over-year changes by lag

The finding

The federal funds rate leads the credit-card delinquency rate, and the lead is long: the correlation between their year-over-year changes peaks at **+8 quarters (r = +0.42)** and stays flat across **7–10 quarters**. The relationship runs one way. Reversed — delinquency leading rates — the correlation never exceeds +0.13 and is negative at every short lead.

Two things about that number matter more than the number itself.

Levels tell you nothing. The contemporaneous correlation of the two series in levels is **+0.21**. Both series trend, and a risk model built on the levels would find almost no signal and conclude there is none. The signal is in the changes, and only after you shift one series by roughly two years.

The lag is a plateau, not a point. Lags of 7, 8, 9 and 10 quarters all fall within 0.03 of the peak. "Eight quarters" is a convenient label for a diffuse response, not a dated event, and a forecast that keys on exactly 8 is claiming precision the data does not carry.

The effect is real in direction and timing but **modest in magnitude** — r ≈ 0.42 accounts for roughly 17% of the variance in delinquency changes. Policy rates set the direction and rough timing of the credit cycle. They do not set its level, and a rate-only loss model will be wrong about how bad it gets.

Results

Analysis run 2026-08-27 on FRED series `FEDFUNDS` (monthly, averaged into quarters) and `DRCCLACBS` (quarterly), 2000Q1–2026Q2, 106 aligned quarters.

Cross-correlation of year-over-year changes

MEASURE	PEAK LAG	R	95% CI	N
YoY changes (overlapping)	+8q	+0.416	+0.01 … +0.70	94 (n/4 = 24 effective)
QoQ changes (non-overlapping)	+9q	+0.294	+0.10 … +0.47	96
YoY changes, excluding 2020–21	+9q	+0.559	+0.39 … +0.69	85
Reverse: delinquency leads rates	—	max +0.13	—	—

The non-overlapping quarter-over-quarter test is the honest significance check — year-over-year differences on quarterly data overlap four-fold, which inflates any naive confidence interval. It agrees on the lag and clears zero on its own.

Turning points, cycle by cycle

RATE PEAK	NEXT DELINQUENCY PEAK	GAP
2007Q1 · 5.26%	2009Q2 · 6.77%	+9 quarters

RATE PEAK	NEXT DELINQUENCY PEAK	GAP
2019Q1 · 2.40%	2020Q1 · 2.69%	+4 quarters
2023Q4 · 5.33%	2024Q2 · 3.22%	+2 quarters

Consistent in sign across three cycles, variable in size. The gaps are shorter than the 8-quarter cross-correlation lag because the *rate of change* in rates peaks well before the rate *level* does — mid-hiking-cycle, one to five quarters ahead of the level peak in these three episodes.

Where the caveats are, stated plainly

- **Pandemic transfers break the relationship outright.** Through 2020–21, delinquency fell while rates fell, the opposite of the pattern. Excluding those eight quarters lifts the peak correlation from +0.42 to +0.56. Direct fiscal support to consumers decouples the series entirely.
- **One cycle in three does not fit.** In 2015–18 the delinquency change peaked *six quarters before* the rate change did. Rates were rising off zero while delinquency rose off a post-crisis floor for its own reasons.
- **The short-lag reading in recent data is not evidence.** Restricted to 2010 onward the correlation appears to peak at 2 quarters (r = +0.75), and post-2022 at 2 quarters (r = +0.99). Neither is a dated lag: 53% of all lags from 0 to 16 quarters exceed |r| = 0.5 in that window and the sign flips strongly negative by 11 quarters, which is the signature of one shared hump. The post-2022 window holds 14 overlapping observations — about 3.5 effective. No lag can be identified from it.

Where the cycle stands now. Rates peaked at 5.33% in 2023Q4, ten quarters ago, and have eased to 3.63%. Delinquency peaked at 3.22% in 2024Q2 and has fallen for eight consecutive quarters to 2.85%. The tightening cycle's transmission has largely run through. The easing that began in 2024 sits inside the 7–10 quarter window that lands in 2026Q3–2027Q2.

What follows for a risk team

1. **Carry the policy rate path as a two-year-ahead input to loss forecasting, not a coincident one.** Book it as a 7–10 quarter band. A model that reads rates and delinquency in the same quarter is reading noise.
2. **Do not build a rate-only loss model.** At ~17% of variance explained, the rate path tells you which way the cycle is turning and roughly when. It does not tell you how far. Unemployment, credit-box policy and balance growth carry the rest, and they are what set the level of loss.
3. **Base case through 2027: continued downward pressure on card delinquency,** from the 1.7pp of easing since 2024Q1 working through the 7–10 quarter window. Carry it with the band, and only

absent a labour-market shock — the lag relationship says nothing about shocks that originate outside the rate path.

4. **Add an explicit override for direct-to-consumer fiscal action.** It is the one intervention in this sample that reversed the sign of the relationship, and any rate-conditioned forecast will be confidently wrong through such an episode.
5. **Re-run on FRED's schedule, and expect the newest quarter to be rate-only.** `DRCCLACBS` publishes roughly a quarter behind `FEDFUNDS` — at retrieval, rates ran to 2026Q3 and delinquency to 2026Q2.

Data: Federal Reserve Bank of St. Louis (FRED). `FEDFUNDS` *— Federal Funds Effective Rate, 319 monthly observations, FRED-updated 2026-08-03.* `DRCCLACBS` *— Delinquency Rate on Credit Card Loans, All Commercial Banks, 106 quarterly observations, FRED-updated 2026-08-25. Retrieved 2026-08-27. Both series are public. Analysis and chart are this run's own output.*
